

EXISTING SYSTEM

- In the current scenario, most users rely on **multiple platforms** or sources for planning their travel, such as:
 - Separate websites for booking hotels
 - Search engines or apps for finding tourist spots
 - Public transport websites or apps for bus/train routes
- There is **no single unified platform** that allows users to:
 - Register and enter their travel details-
 - Get customized recommendations based on their travel purpose (exams, internships, business trips)
 - Receive hotel, tourist spot, and route suggestions in one place
- **Manual effort** is often required by users to search and compare information from different sources, which is:
 - Time-consuming
 - Confusing for people unfamiliar with the location
 - Prone to missing out on better options or accurate route details
- These existing systems lack **integration between location selection and accommodation or transport suggestions**, resulting in:
 - Poor user experience
 - Incomplete travel guidance
 - Limited personalization based on stay duration or center location

PROPOSED SYSTEM

- The proposed system is a **Travel Guidance Web Application** that provides a complete solution for users traveling to unfamiliar locations for exams, internships, or business purposes.
- It allows users to **register** and enter essential travel details such as country, state, district, duration of stay, and center location in a single platform.
- Based on the selected **center location**, the system provides:
 - **Nearby hotel suggestions** for accommodation
 - **Tourist spot recommendations** within the surrounding area
 - **Travel route guidance** (train route or bus route number) based on the user's chosen mode of transport
- The system is designed to **reduce manual effort** by integrating all the necessary information into one easy-to-use web platform.
- It uses a **MySQL database** to store user data and travel preferences, ensuring organized and secure data handling.
- The **backend logic is implemented in Python using Flask**, which efficiently manages database connections and processes user inputs.
- The **frontend is developed using HTML and CSS**, offering a clean and responsive interface for a better user experience.
- The system is flexible, scalable, and designed to improve **planning efficiency, accuracy, and user convenience**.

KEY FEATURES OF THE PROJECT

- **USER REGISTRATION:**
Allows users to register by entering their name, mobile number, and email ID to begin planning their travel.
- **LOCATION-BASED TRAVEL PLANNING:**
Users can select their country, state, and district to personalize their travel plan.
- **DURATION SELECTION:**
Users can specify the number of days they plan to stay, which helps tailor hotel and tourist spot recommendations.
- **CENTER LOCATION SELECTION:**
Enables the user to choose a center location such as an exam venue, internship site, or business meeting place.
- **HOTEL SUGGESTIONS:**
The system suggests nearby hotels for accommodation based on the selected center location.
- **NEARBY TOURIST SPOT RECOMMENDATIONS:**
Recommends tourist attractions located close to the center location to help users make the most of their stay.
- **TRAVEL ROUTE GUIDANCE:**
Offers appropriate travel route suggestions based on the user's selected mode of transportation:
 - Train route information
 - Bus route numbers
- **DATA STORAGE:**
All user inputs and travel details are securely stored in a MySQL database for future reference or analysis.
- **BACKEND PROCESSING:**
Python with Flask is used to manage backend operations, handle form submissions, and connect to the database.
- **USER-FRIENDLY INTERFACE:**
The front-end is developed using HTML and CSS, providing a clean and responsive user experience.
- **INTEGRATED WEB SOLUTION:**
Combines frontend, backend, and database technologies to deliver a seamless and functional web application.

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TOOLS AND TECHNOLOGIES USED

FRONTEND TECHNOLOGIES:

- **HTML** – Used to create the structure of the web pages.
- **CSS** – Used for styling the user interface and enhancing the visual appearance of the website.

BACKEND TECHNOLOGY:

- **Python** – Used for backend logic to process user data and connect with the database.

WEB FRAMEWORK:

- **Flask** – A lightweight Python web framework used to build and manage the web application routes and server-side functionalities.

DATABASE:

- **MySQL** – Used for storing user registration details, travel preferences, hotel data, tourist spot information, and transport routes.

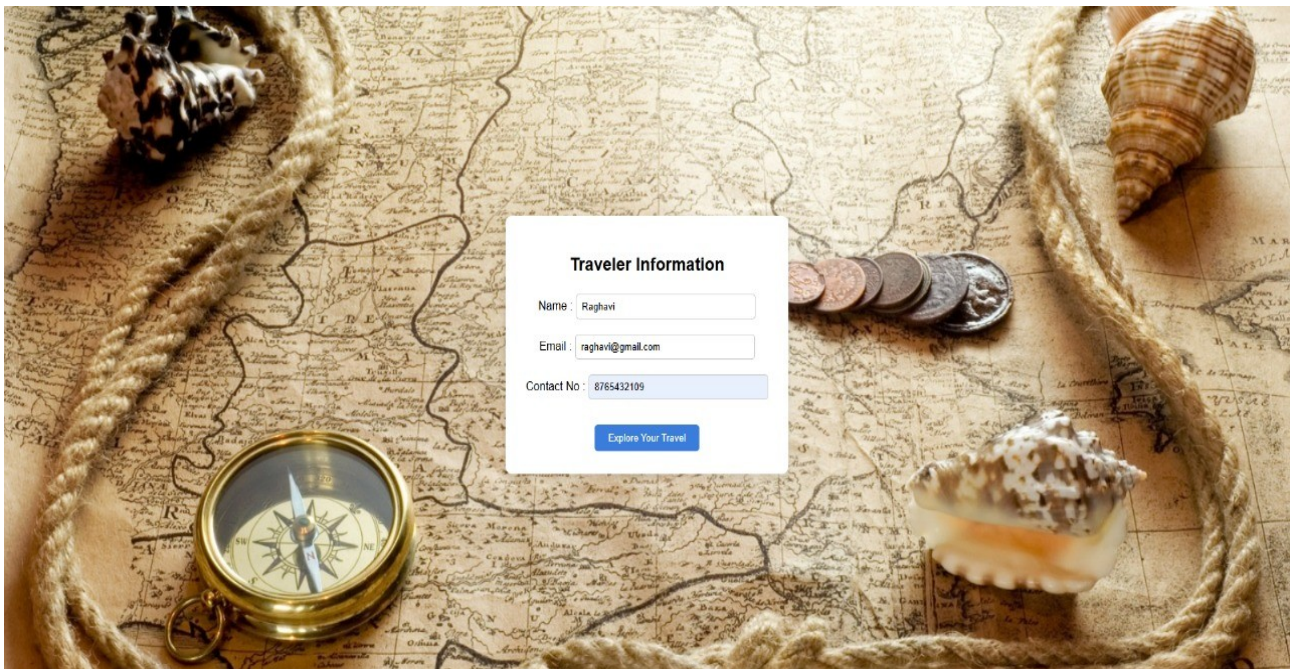
DEVELOPMENT TOOLS:

- **Python IDLE** – Used for writing and testing Python code.
- **MySQL Workbench / Command Line Interface** – Used for creating and managing the database and tables.
- **Web Browser** – Used for testing and running the web application.

EXPLANATION OF EACH PROCESS :

USER REGISTRATION

The user registration form collects essential details such as the **Name**, **Mobile Number**, and **Email ID** of the user. Once the information is submitted through the **Submit** button, the data is stored and the user is redirected to the travel planning form. This serves as the entry point into the travel guidance system.



TRAVEL PLANNING FORM

In the travel planning form, the user is guided through a **step-by-step location selection** process. It begins with selecting the **Country** from a dropdown menu. Based on the selected country, the **State dropdown** is populated dynamically. Similarly, the **District dropdown** is updated according to the selected state. The user must also enter the **number of travel days**, which is restricted to either **1 or 2 days**, and provide a **center location** such as a college, exam center, or company around which the travel recommendations will be based.



Select State:

Tamil Nadu

Select District:

Madurai

Select Exam Center:

Thiagarajar Collgo

Select Hotel:

ITC Maurya

Select Tourist Place:

Gandhi Museum

Route from Exam Center to Hotel:
Bus: APSRTC Bus 145 (VSU Campus → Nellore City) → Auto
Train: Not available
Car: VRC Center Road → Taj Krishna (~13 km, ~25 mins)

Route from Exam Center to Tourist Place:
Bus: Bus from NGGO Colony → Peelamedu → Prozone. Train: Not available — prefer bus or car. Car: NGGO Colony → Avinashi Rd (~11 km, ~25 mins)

Feedback:

excellent

Submit Travel Plan



STATE SELECTION

The user begins the travel planning process by selecting a state. This selection acts as the first level of filtering and determines the list of available districts. It helps narrow down the search area for exam centers and related travel options. All subsequent suggestions are based on the selected state.



Plan Your Travel

Select State:

-- Select State --

Select D

Himachal Pradesh

Jharkhand

Karnataka

Select E

Kerala

Madhya Pradesh

Maharashtra

Select H

Manipur

Meghalaya

Mizoram

Select T

Nagaland

Odisha

Route fr

Punjab

Route fr

Rajasthan

Sikkim

Feedback

Tamil Nadu

Telangana

Tripura

Uttar Pradesh

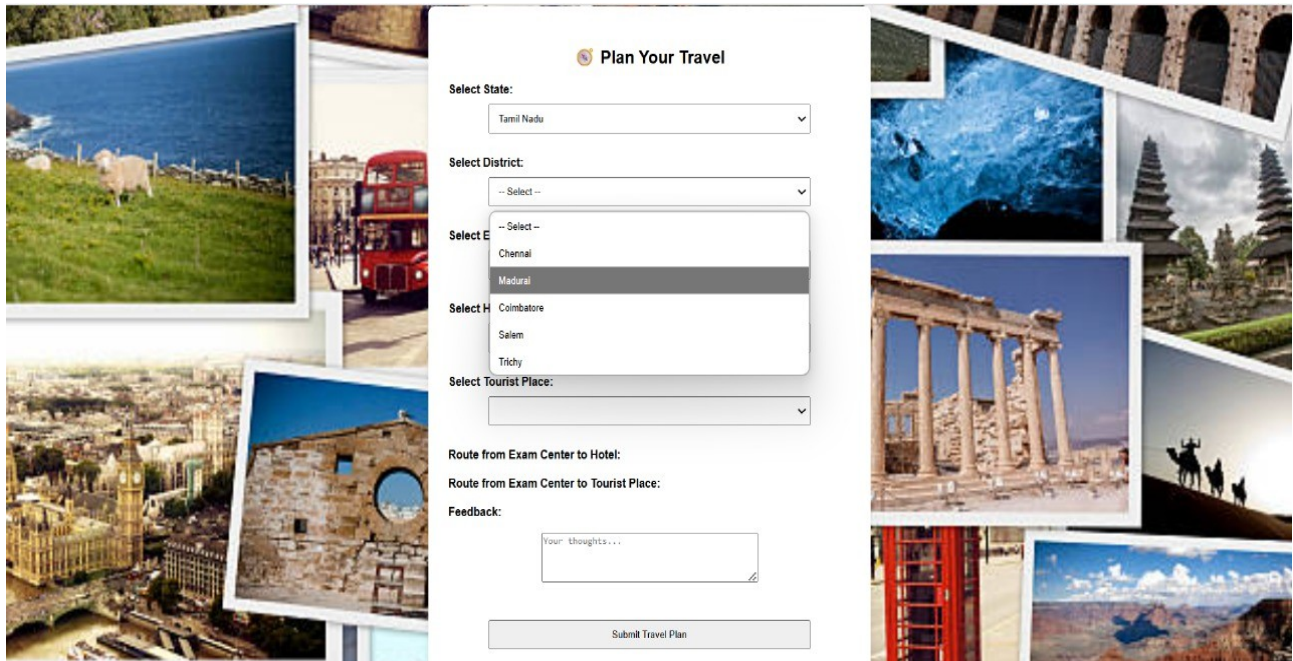
Uttarakhand

West Bengal



DISTRICT SELECTION

Once a state is selected, the system dynamically updates and displays the **districts** that fall under that state. This helps narrow down the location further and ensures relevant exam center listings.



Plan Your Travel

Select State:
Tamil Nadu

Select District:
-- Select --

Select Exam Center:
-- Select --
Chennai
Madurai
Coimbatore
Salem
Tiruchy

Select Tourist Place:
-- Select --

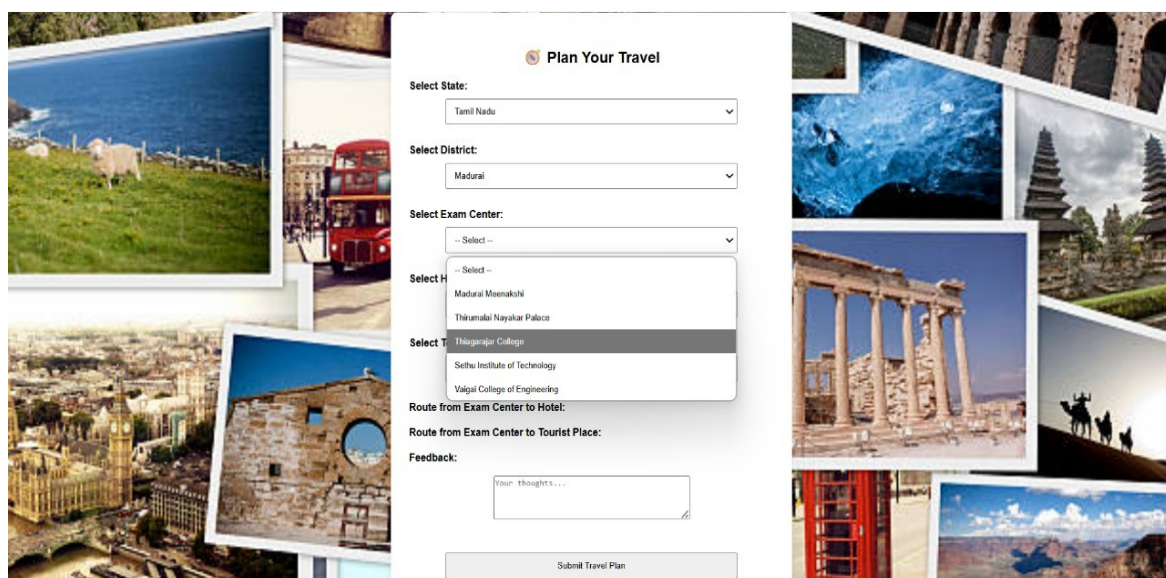
Route from Exam Center to Hotel:
Route from Exam Center to Tourist Place:

Feedback:
Your thoughts...

Submit Travel Plan

CENTER LOCATION

The **center location** is a key input in the travel planning form, as it helps to determine travel recommendations. The user is prompted to specify a **center location**, which could be a significant place like a **college**, **exam center**, or **company**. The system uses this information to suggest **nearby hotels**, **tourist spots**, and suitable **routes**. The center location serves as a reference point for tailoring the travel recommendations, ensuring that all suggestions are relevant to the user's specific needs and travel goals.



Plan Your Travel

Select State:
Tamil Nadu

Select District:
Madurai

Select Exam Center:
-- Select --

Select Hotel:
-- Select --
Madurai Moenakudi
Thirumalai Nayakar Palace

Select Tourist Place:
Thiruvannamalai College
Selva Institute of Technology
Vadga College of Engineering

Route from Exam Center to Hotel:
Route from Exam Center to Tourist Place:

Feedback:
Your thoughts...

Submit Travel Plan

HOTEL SUGGESTION

Once the center location is entered, the system fetches and displays a list of **nearby hotels**. The user can browse through these suggestions and choose one according to their preference. This hotel selection helps in narrowing down nearby tourist spots and available routes.

Plan Your Travel

Select State:

Select District:

Select Exam Center:

Select Hotel:

Route from Exam Center to Hotel:

Route from Exam Center to Tourist Place:

Feedback:

TOURIST SPOT RECOMMENDATION

After the hotel is selected, the application displays **nearby tourist spots** related to the hotel or center location. Each tourist spot is shown with its **Name**, **Type** (such as temple, beach, museum, etc.), and **Distance** from the selected center or hotel. This provides the user with an overview of what they can explore during their stay.

Plan Your Travel

Select State:

Select District:

Select Exam Center:

Select Hotel:

Select Tourist Place:

Route from Exam Center to Hotel:

Route from Exam Center to Tourist Place:

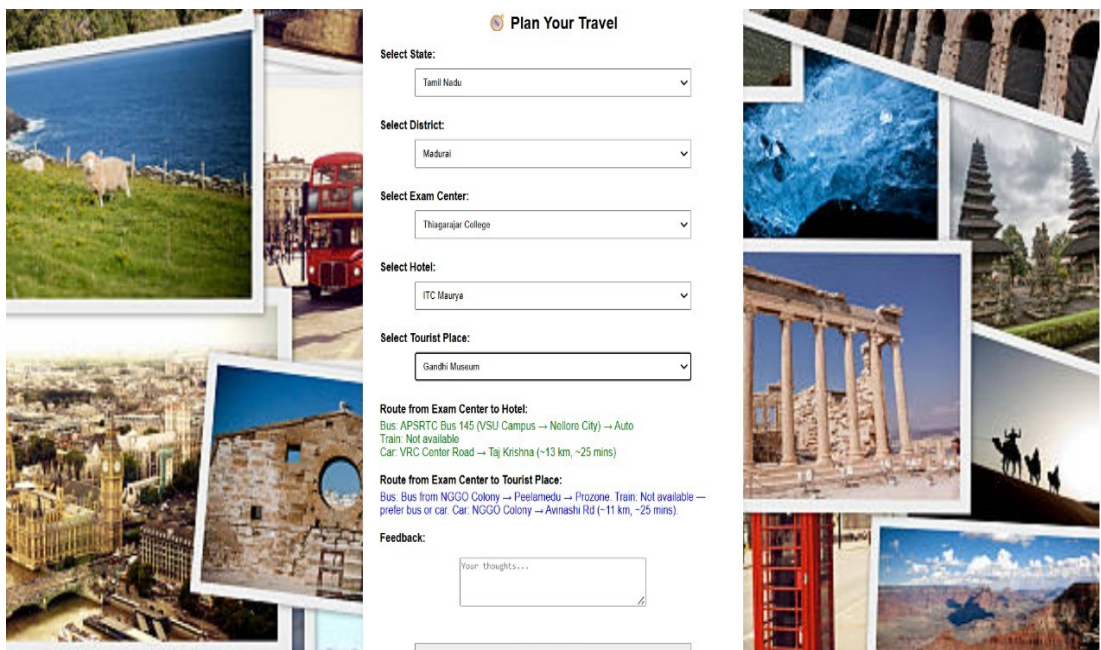
Feedback:

ROUTE SUGGESTION

After selecting a hotel or tourist spot, the system suggests the best possible routes from the exam center to the chosen destination. It intelligently analyzes available transportation options and presents a clear route overview. This includes:

- **Local Train:** If a local train is available between the locations, the system highlights the nearest stations, train timings, and estimated travel duration.
- **Bus:** When a bus route is available, the system shows nearby bus stops, route numbers, frequency, and approximate travel time.
- **Car/Private Vehicle:** For users preferring to travel by car or taxi, the system displays the road route, total distance in kilometers, estimated driving time, and possible traffic conditions.

Each route option is presented with key information, allowing the user to compare and choose the most convenient travel method. This ensures a smooth and well-informed travel experience.



Plan Your Travel

Select State:

Select District:

Select Exam Center:

Select Hotel:

Select Tourist Place:

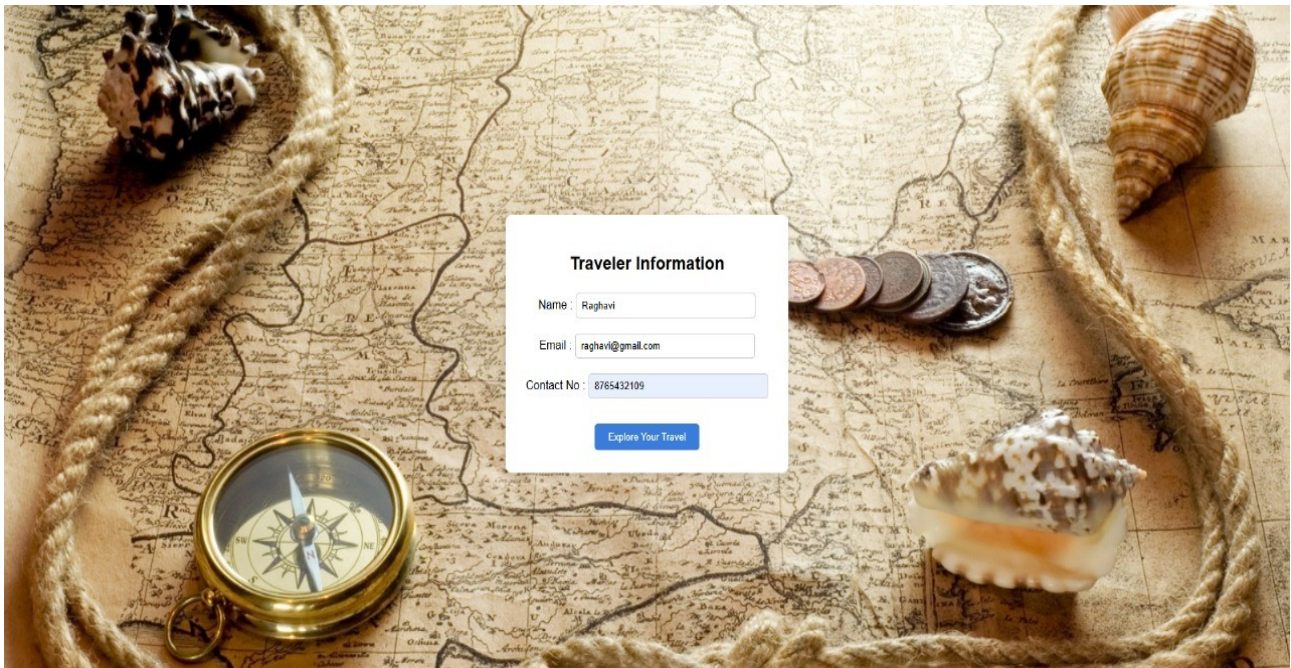
Route from Exam Center to Hotel:
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Train: Not available
Car: VRC Center Road → Taj Krishna (~13 km, ~25 mins)

Route from Exam Center to Tourist Place:
Bus: Bus from NGGO Colony → Peelamedu → Prozone. Train: Not available — prefer bus or car. Car: NGGO Colony → Avinashi Rd (~11 km, ~25 mins).

Feedback:

FINAL OVERVIEW :

Before completing the process, the application presents a **final summary** of all the inputs and choices made by the user. This includes the user's personal details, selected **Country, State, District**, specified **Center Location, Chosen Hotel**, nearby **Tourist Places**, and the selected **Route Information**. On clicking the **Final Submit** button, all of this data is stored securely in the **MySQL database**, concluding the travel planning process.



Select State:

Select District:

Select Exam Center:

Select Hotel:

Select Tourist Place:

Route from Exam Center to Hotel:
 Bus: APSRTC Bus 145 (VSU Campus → Nellore City) → Auto
 Train: Not available
 Car: VRC Center Road → Taj Krishna (~13 km, ~25 mins)

Route from Exam Center to Tourist Place:
 Bus: Bus from NGGO Colony → Peelamedu → Prozone. Train: Not available —
 prefer bus or car. Car: NGGO Colony → Avinashi Rd (~11 km, ~25 mins)

Feedback:



DATABASE TABLES USED :

The **Travel Guidance Web Application** utilizes a well-structured relational database named `travel_db` in **MySQL** to store and manage all the necessary information related to the user's travel planning. The database consists of the following **eight tables**:

1. **user_info**: This table stores the registration details of users, such as name, mobile number, and email ID.
2. **travel_plan**: This table holds information related to the user's travel plan, including the selected country, state, district, number of travel days, center location, chosen hotel, tourist places, and route details.
3. **states**: This table contains the list of states, usually linked to a corresponding country.
4. **districts**: It holds the list of districts and is connected to the `states` table to enable dynamic population based on the selected state.
5. **district_state**: This is a reference table that maps districts to their corresponding states to ensure relational integrity during location selection.
6. **exam_centers**: This table includes predefined central locations like colleges, companies, or exam centers around which hotel and tourist spot suggestions are provided.
7. **hotels**: This table contains data about hotels, such as hotel name, location, and other details needed for hotel recommendation.
8. **hotels_centers**: This mapping table connects hotels with nearby exam centers or central locations, helping the system suggest the most relevant hotels based on the selected center.

Together, these tables form the backbone of the application's database and enable smooth functionality for user interaction, travel planning, and data storage.

```
mysql> show tables;
+-----+
| Tables_in_travel_db |
+-----+
| districts             |
| exam_centers          |
| exam_places           |
| hotels                |
| states                |
| tourist_places        |
| travel_plan           |
| user_information      |
+-----+
8 rows in set (0.00 sec)
```

TABLE: user_information

Stores basic information about registered users.

- **id**: Unique identifier for each user (Primary Key).
- **name**: Name of the user.
- **email**: User's email address.
- **phone**: User's contact number.
- **registered_at**: Timestamp indicating when the user registered.

```
mysql> desc user_information;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
name	varchar(100)	YES		NULL	
email	varchar(100)	YES		NULL	
phone	varchar(15)	YES		NULL	
registered_at	datetime	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

5 rows in set (0.00 sec)

TABLE: travel_plan

Records the travel plan generated by users based on selected inputs.

- **id**: Unique identifier for each travel plan (Primary Key).
- **state**: Name of the selected state.
- **district**: Name of the selected district.
- **center**: Name of the selected exam center.
- **hotel**: Name of the selected hotel.
- **tourist_place**: Name of the selected tourist place.
- **center_to_hotel**: Route details from the exam center to the hotel.
- **center_to_touristplace**: Route details from the exam center to the tourist place.
- **feedback**: User feedback or additional notes.


```
mysql> desc travel_plan;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
state	varchar(50)	YES		NULL	
district	varchar(50)	YES		NULL	
center	varchar(100)	YES		NULL	
hotel	varchar(100)	YES		NULL	
tourist_place	varchar(100)	YES		NULL	
center_to_hotel	text	YES		NULL	
center_to_touristplace	text	YES		NULL	
feedback	text	YES		NULL	

```
9 rows in set (0.00 sec)
```

TABLE: states

Stores information about different states available in the travel planning system.

- **state_id**: Unique identifier for each state (Primary Key).
- **state_name**: Name of the state (e.g., Tamil Nadu, Maharashtra).

```
mysql> desc states;
```

Field	Type	Null	Key	Default	Extra
state_id	int	NO	PRI	NULL	auto_increment
state_name	varchar(100)	NO		NULL	

```
2 rows in set (0.00 sec)
```

TABLE: districts

Stores districts that belong to specific states.

- **district_id**: Unique identifier for each district (Primary Key).
- **district_name**: Name of the district.
- **state_id**: Foreign key referencing the state this district belongs to.

```
mysql> desc districts;
```

Field	Type	Null	Key	Default	Extra
district_id	int	NO	PRI	NULL	auto_increment
district_name	varchar(100)	NO		NULL	
state_id	int	YES	MUL	NULL	

```
3 rows in set (0.01 sec)
```

TABLE: exam_centers

Contains data about various exam centers located in different districts.

- **center_id**: Unique identifier for each exam center (Primary Key).
- **center_name**: Name of the exam center.
- **district_id**: Foreign key referencing the associated district.

```
mysql> desc exam_centers;
```

Field	Type	Null	Key	Default	Extra
center_id	int	NO	PRI	NULL	auto_increment
center_name	varchar(150)	NO		NULL	
district_id	int	YES	MUL	NULL	

3 rows in set (0.00 sec)

TABLE: hotels

Holds details of hotels available near specific exam centers.

- **hotel_id**: Unique identifier for each hotel (Primary Key).
- **hotel_name**: Name of the hotel.
- **center_id**: Foreign key linking the hotel to its nearby exam center.
- **hotel_examcenter_route**: Description or details of the route from the exam center to the hotel.

```
mysql> desc hotels;
```

Field	Type	Null	Key	Default	Extra
hotel_id	int	NO	PRI	NULL	auto_increment
hotel_name	varchar(150)	NO		NULL	
center_id	int	YES	MUL	NULL	
hotel_examcenter_route	text	YES		NULL	

4 rows in set (0.00 sec)

TABLE: tourist_places

Stores tourist attractions near the exam centers.

- **place_id**: Unique identifier for each tourist place (Primary Key).
- **place_name**: Name of the tourist place.
- **description**: Brief overview or details about the place.
- **center_id**: Foreign key linking the tourist spot to a nearby exam center.
- **route**: Details of the route from the exam center to the tourist place.

```
mysql> desc tourist_places;
```

Field	Type	Null	Key	Default	Extra
place_id	int	NO	PRI	NULL	auto_increment
place_name	varchar(100)	YES		NULL	
description	text	YES		NULL	
center_id	int	YES	MUL	NULL	
route	text	YES		NULL	

```
5 rows in set (0.00 sec)
```